

## MEMBERS' NAME

## MATIC NUMBERS

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## QUESTION 1

$$(a) (A \cap B) \cup (B \cap C) \subseteq B$$

$$A \cap B = \{3, 5\}$$

$$B \cap C = \{5, 6\}$$

$$(A \cap B) \cup (B \cap C) = \{3, 5, 6\}$$

Subset of  $(A \cap B) \cup (B \cap C)$  $\emptyset, \{3\}, \{5\}, \{6\}, \{3, 5\}, \{3, 6\}, \{5, 6\}, \{3, 5, 6\}$  $\therefore (A \cap B) \cup (B \cap C)$  is a subset of  $B$ .

$$(b) (A \cup C) - B = (A - B) \cup (C - B)$$

RHS :

$$= (A \cap B') \cup (A \cap B')$$

$$= (A \cup C) \cap B'$$

$$= (A \cup C) - B$$

LHS :

$$= (A \cup C) \cap B' \quad (A \cup C)' \cap B$$

$$= (A \cap B') \cup (C \cap B') \quad (A' \cup C') \cap B$$

$$= (A - B) \cup (C - B) \quad \therefore \text{proven}$$

$$(A' \cap B) \cup (C' \cap B)$$

$$(A - B) \cup (C - B)$$

 $\therefore \text{proven}$



$$(c) \quad A \oplus (B \cap C), (A \oplus B) \oplus C$$

$$\begin{aligned} A \oplus (B \cap C) &= \{1, 3, 5, 7\} \oplus \{5, 6\} \\ &= \{1, 3, 6, 7\} \end{aligned}$$

$$(A \oplus B) \oplus C :$$

$$\begin{aligned} (A \oplus B) &= \{1, 3, 5, 7\} \oplus \{3, 4, 5, 6\} \\ &= \{1, 4, 6, 7\} \end{aligned}$$

$$\begin{aligned} (A \oplus B) \oplus C &= \{1, 4, 6, 7\} \oplus \{5, 6, 7, 8\} \\ &= \{1, 4, 5, 8\} \end{aligned}$$

$$\therefore A \oplus (B \cap C) \neq (A \oplus B) \oplus C$$



## QUESTION 2

$$P = \{2, 3, 5, 7, 11, 13, 17, 19\}$$

$$E = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$$

$$D = \{1, 2, 3, 4, 6, 9, 12, 18\}$$

(a) "all elements of  $P$  that are not in  $D$  are also in  $E$ "

$$\forall x \in P, (x \notin D) \rightarrow (x \in E)$$

(b)  $P - D = \{5, 7, 11, 13, 17, 19\}$

$$\{5, 7, 11, 13, 17, 19\} \notin E$$

the element  $\{5, 7, 11, 13, 17, 19\}$  is not

$\therefore$  The statement is disproven because  $\neg (P \setminus D) \rightarrow (x \in E)$

(c)  $(P \cap E) \cup D = D$

$$P \cap E = \{2\}$$

$$D = \{1, 2, 3, 4, 6, 9, 12, 18\}$$

$$(P \cap E) \cup D =$$

$$= \{2\} \cup \{1, 2, 3, 4, 6, 9, 12, 18\}$$

$$= \{1, 2, 3, 4, 6, 9, 12, 18\}$$

$$= (P \cap E) \cup D \quad \therefore \text{proven}$$



### \* QUESTION 3

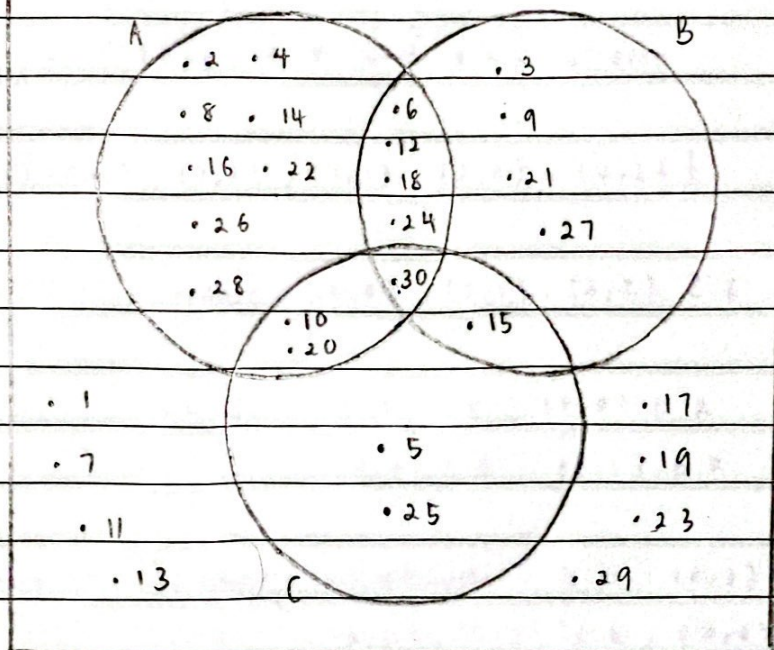
$$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30\}$$

$$B = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30\}$$

$$C = \{5, 10, 15, 20, 25, 30\}$$

(a)

U



(b) number of elements in each region

$$i) |A \cap B \cap C| = 1$$

$$ii) |A - B - C| = 8$$

$$iii) |B - C - A| = 4$$

$$iv) |C - A - B| = 2$$

$$v) |A \cap B| = 4$$

$$vi) |A \cap C| = 2$$

$$vii) |B \cap C| = 1$$

$$viii) |(A \cup B \cup C)'| = 8$$



## QUESTION 4

$$(a) \quad X = \{0, 1, 2\} \quad Y = \{a, b\}$$

$$X \times Y = \{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b)\}$$

$$(b) \quad X \times X = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)\}$$

$$R \subseteq X \times X, (x, y) \in R \iff x + y \text{ is even}$$

$$R = \{(0, 0), (0, 2), (1, 1), (2, 0), (2, 2)\}$$

(c) Reflexive because  $(0, 0), (1, 1), (2, 2) \in R$

Symmetric because  $(0, 1) \in R, (1, 0) \in R$

$(0, 2) \in R, (2, 0) \in R$

Transitive because  $(0, 0), (0, 2) \in R, (0, 2) \in R$

$(0, 2), (2, 0) \in R, (0, 0) \in R$

$(0, 2), (2, 2) \in R, (0, 2) \in R$

$(2, 0), (0, 0) \in R, (2, 0) \in R$

$(2, 0), (0, 2) \in R, (2, 2) \in R$

$(2, 2), (2, 0) \in R, (2, 0) \in R$



# QUESTION 5

(a) contrapositive : "If  $x$  is not a multiple of 4, then  $x$  is not even or  $x^2$  is not even."

let  $x \neq$  multiple of 4

$$x = 4y + 1, \text{ where } y \in \mathbb{Z}$$

$x$  will be an odd number,  $x^2$  will also be odd. [proven]

$\therefore$  The contrapositive "If  $x$  is not a multiple of 4, then  $x$  is not even or  $x^2$  is not even" is proven.

(b) Assume the conclusion is false, then  $m$  is odd or  $n$  is odd

$$m = 2x + 1, \quad y = 2y + 1$$

$$mn = (2x + 1)(2y + 1)$$

$$= 4xy + 2x + 2y + 1$$

$$= 2(2xy + x + y) + 1 \quad [\text{odd}]$$

$\therefore$  The statement "For all integers  $m$  and  $n$ , if  $mn$  is even, then  $m$  is even or  $n$  is even." is true.

(c) Assume square root of 3 is rational

$$\sqrt{3} = \frac{a}{b}$$

$a^2$  is a multiple of 3

$\therefore \sqrt{3}$  is irrational

$a = 3k$  for some integer  $k$

[proven]

$$3 = \frac{a^2}{b^2}$$

$$(3k)^2 = 3b^2$$

$$9k^2 = 3b^2$$

$$a^2 = 3b^2$$

$$b^2 = 3k^2$$

so,  $b^2$  is also a multiple of 3.



## QUESTION 6

$$(a) f(x) = 3x + 2$$

$$3x_1 + 2 = 3x_2 + 2$$

$$3x_1 = 3x_2$$

$$x_1 = x_2$$

$\therefore$  proven

$$(b) y = 3x + 2$$

$$3x + 2 = y$$

$$x = \frac{y-2}{3}$$

$$f(x) = f\left(\frac{y-2}{3}\right)$$

$$= 3\left(\frac{y-2}{3}\right) + 2$$

$$= y - 2 + 2$$

$$f(x) = y \quad \therefore \text{proven}$$

$$(c) f^{-1}(x) = y$$

$$y = 3x + 2$$

$$\frac{y-2}{3} = x$$

$\therefore$

$$f^{-1}(x) = \frac{x-2}{3}$$



## QUESTION 7

$$(a) \quad g(x) = x^2, \quad h(x) = x+1$$

$$\begin{aligned} (g \circ h)(x) &= (x+1)^2 \\ &= (x+1)(x+1) \\ &= x^2 + 2x + 1 \end{aligned}$$

$$(h \circ g)(x) = x^2 + 1$$

$$\begin{aligned} (b) \quad g \circ hg(x) &= (x^2 + 1)^2 \\ &= x^4 + 2x^2 + 1 \end{aligned}$$

$$(x_1^2 + 1)^2 = (x_2^2 + 1)^2$$

$$x_1^2 + 1 = \pm (x_2^2 + 1)$$

$\therefore$  not one-to-one function because it cannot produce negative value

$$g \circ hg(0) = (0^2 + 1)^2 = 1 \quad \therefore \text{not onto}$$

$$\begin{aligned} h \circ gh(x) &= (x+1)^2 + 1 \\ &= x^2 + 2x + 2 \end{aligned}$$

$$x_1^2 + 2x_1 + 2 = x_2^2 + 2x_2 + 2$$

$$x_1^2 + 2x_1 = x_2^2 + 2x_2$$

$$x_1 = x_2$$

$$x_1 + x_2 + 2 = 0, \text{ or } x = -x_1 - 2 \quad \therefore \text{not one-to-one function}$$

$$h \circ gh(-1) = (-1)^2 + 2(-1) + 2 = 1 \quad \therefore \text{not onto}$$

$\therefore g \circ hg(x)$  is neither one-to-one nor onto

$\therefore h \circ gh(x)$  is neither one-to-one nor onto

$$(c) \quad (h \circ g)^{-1}(x) = y \quad x = \sqrt{y-1}$$

$$y = x^2 + 1$$

$$y - 1 = x^2$$

$$\therefore (h \circ g)^{-1}(x) = \pm \sqrt{y-1}$$

$$y \geq 1, \quad y-1 \geq 0$$